

Supporting Information

Time-to-Event Estimation with Unreliably Reported Events in Medicare Health Plan Payment

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1. Additional information about simulated data

Table 1: **Hierarchical Condition Categories (HCCs) included in the Medicare Advantage risk adjustment algorithm version 28 (V28).** Each of the HCCs in this table are described as written in the Centers for Medicare and Medicaid Services' (CMS) 2024 Medicare Advantage Advance Notice¹. Competing events, if applicable, are also noted from the 2024 CMS Risk Adjustment Model Software and ICD-10 Mappings, specifically the Midyear/Final Model Software, which is in the SAS language². If a set of competing events is not N/A, then, if the HCC in that row is coded, none of the competing event HCCs can also be coded for billing purposes. The hierarchy described in the "Competing Events" column can also be found in the upcoding R package.

HCC	HCC Description	Competing Events
HCC1	HIV/AIDS	N/A
HCC2	Septicemia, sepsis, systemic inflammatory response syndrome/shock	N/A
HCC6	Opportunistic infections	N/A
HCC17	Cancer metastatic to lung, liver, brain, and other organs; acute myeloid leukemia except promyelocytic	HCC18, HCC19, HCC20, HCC21, HCC22, HCC23
HCC18	Cancer metastatic to bone, other and unspecified metastatic cancer; acute leukemia except myeloid	HCC19, HCC20, HCC21, HCC22, HCC23
HCC19	Myelodysplastic syndromes, multiple myeloma, and other cancers	HCC20, HCC21, HCC22, HCC23
HCC20	Lung and other severe cancers	HCC21, HCC22, HCC23

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HCC	HCC Description	Competing Events
HCC21	Lymphoma and other cancers	HCC22, HCC23
HCC22	Bladder, colorectal, and other cancers	HCC23
HCC23	Prostate, breast, and other cancers and tumors	N/A
HCC35	Pancreas transplant status	HCC36, HCC37, HCC38
HCC36	Diabetes with severe acute complications	HCC37, HCC38
HCC37	Diabetes with chronic complications	HCC38
HCC38	Diabetes with glycemic, unspecified, or no complications	N/A
HCC48	Morbid obesity	N/A
HCC49	Specified lysosomal storage disorders	N/A
HCC50	Amyloidosis, porphyria, and other specified metabolic disorders	N/A
HCC51	Addison's and Cushing's diseases, acromegaly, and other specified endocrine disorders	N/A
HCC62	Liver transplant status/complications	HCC63, HCC64, HCC65, HCC68
HCC63	Chronic liver failure/end-stage liver disorders	HCC64, HCC65, HCC68, HCC202
HCC64	Cirrhosis of liver	HCC65, HCC68
HCC65	Chronic hepatitis	N/A
HCC68	Cholangitis and obstruction of bile duct without gallstones	N/A
HCC77	Intestine transplant status/complications	HCC78, HCC80, HCC81
HCC78	Intestinal obstruction/perforation	N/A
HCC79	Chronic pancreatitis	N/A
HCC80	Crohn's disease (regional enteritis)	HCC81
HCC81	Ulcerative colitis	N/A
HCC92	Bone/joint/muscle/severe soft tissue infections/necrosis	N/A

HCC	HCC Description	Competing Events
HCC93	Rheumatoid arthritis and other specified inflammatory rheumatic disorders	HCC94
HCC94	Systemic lupus erythematosus and other specified systemic connective tissue disorders	N/A
HCC107	Sickle cell anemia (Hb-SS) and thalassemia beta zero	HCC108
HCC108	Sickle cell disorders, except sickle cell anemia (Hb-SS) and thalassemia beta zero; beta thalassemia major	N/A
HCC109	Acquired hemolytic, aplastic, and sideroblastic anemias	N/A
HCC111	Hemophilia, male	HCC112
HCC112	Immune thrombocytopenia and specified coagulation defects and hemorrhagic conditions	N/A
HCC114	Common variable and combined immunodeficiencies	HCC115
HCC115	Specified immunodeficiencies and white blood cell disorders	N/A
HCC125	Dementia, severe	HCC126, HCC127
HCC126	Dementia, moderate	HCC127
HCC127	Dementia, mild or unspecified	N/A
HCC135	Drug use with psychotic complications	HCC136, HCC137, HCC138, HCC139
HCC136	Alcohol use with psychotic complications	HCC137, HCC138, HCC139
HCC137	Drug use disorder, moderate/severe, or drug use with non-psychotic complications	HCC138, HCC139
HCC138	Drug use disorder, mild, uncomplicated, except cannabis	HCC139
HCC139	Alcohol use disorder, moderate/severe, or alcohol use with specified non-psychotic complications	N/A
HCC151	Schizophrenia	HCC152, HCC153, HCC154, HCC155

HCC	HCC Description	Competing Events
HCC152	Psychosis, except schizophrenia	HCC153, HCC154, HCC155
HCC153	Personality disorders; anorexia/bulimia nervosa	HCC154, HCC155
HCC154	Bipolar disorders without psychosis	HCC155
HCC155	Major depression, moderate or severe, without psychosis	N/A
HCC180	Quadriplegia	HCC181, HCC182, HCC253, HCC254
HCC181	Paraplegia	HCC182, HCC254
HCC182	Spinal cord disorders/injuries	N/A
HCC190	Amyotrophic lateral sclerosis and other motor neuron disease, spinal muscular atrophy	N/A
HCC191	Quadriplegic cerebral palsy	HCC180, HCC181, HCC182, HCC192, HCC253, HCC254
HCC192	Cerebral palsy, except quadriplegic	HCC180, HCC181, HCC182, HCC253, HCC254
HCC193	Chronic inflammatory demyelinating polyneuritis and multifocal motor neuropathy	N/A
HCC195	Myasthenia gravis with (acute) exacerbation	HCC196
HCC196	Myasthenia gravis without (acute) exacerbation and other myoneural disorders	N/A
HCC197	Muscular dystrophy	N/A
HCC198	Multiple sclerosis	N/A
HCC199	Parkinson and other degenerative disease of basal ganglia	N/A
HCC200	Friedreich and other hereditary ataxias; Huntington disease	N/A
HCC201	Seizure disorders and convulsions	N/A
HCC202	Coma, brain compression/anoxic damage	N/A
HCC211	Respirator dependence/tracheostomy status/complications	HCC212, HCC213

HCC	HCC Description	Competing Events
HCC212	Respiratory arrest	HCC213
HCC213	Cardio-respiratory failure and shock	N/A
HCC221	Heart transplant status/complications	HCC222, HCC223, HCC224, HCC225, HCC226, HCC227
HCC222	End stage heart failure	HCC223, HCC224, HCC225, HCC226, HCC227
HCC223	Heart failure with heart assist device/artificial heart	HCC224, HCC225, HCC226, HCC227
HCC224	Acute on chronic heart failure	HCC225, HCC226, HCC227
HCC225	Acute heart failure (excludes acute on chronic)	HCC226, HCC227
HCC226	Heart failure, except end stage and acute	HCC227
HCC227	Cardiomyopathy/myocarditis	N/A
HCC228	Acute myocardial infarction	HCC229
HCC229	Unstable angina and other acute ischemic heart disease	N/A
HCC238	Specified heart arrhythmias	N/A
HCC248	Intracranial hemorrhage	HCC249
HCC249	Ischemic or unspecified stroke	N/A
HCC253	Hemiplegia/hemiparesis	HCC254
HCC254	Monoplegia, other paralytic syndromes	N/A
HCC263	Atherosclerosis of arteries of the extremities with ulceration or gangrene	HCC264, HCC383, HCC409
HCC264	Vascular disease with complications	N/A
HCC267	Deep vein thrombosis and pulmonary embolism	N/A
HCC276	Lung transplant status/complications	HCC277, HCC278, HCC279, HCC280
HCC277	Cystic fibrosis	HCC278, HCC279, HCC280
HCC278	Idiopathic pulmonary fibrosis and lung involvement in systemic sclerosis	HCC279, HCC280
HCC279	Severe persistent asthma	HCC280
HCC280	Chronic obstructive pulmonary disease, interstitial lung disorders, and other chronic lung disorders	N/A

HCC	HCC Description	Competing Events
HCC282	Aspiration and specified bacterial pneumonias	HCC283
HCC283	Empyema, lung abscess	N/A
HCC298	Severe diabetic eye disease, retinal vein occlusion, and vitreous hemorrhage	N/A
HCC300	Exudative macular degeneration	N/A
HCC326	Chronic kidney disease, stage 5	HCC327, HCC328, HCC329
HCC327	Chronic kidney disease, severe (stage 4)	HCC328, HCC329
HCC328	Chronic kidney disease, moderate (stage 3B)	HCC329
HCC329	Chronic kidney disease, moderate (stage 3, except 3B)	N/A
HCC379	Pressure ulcer of skin with necrosis through to muscle, tendon, or bone	HCC380, HCC381, HCC382, HCC383
HCC380	Chronic ulcer of skin, except pressure, through to bone or muscle	HCC381, HCC382, HCC383
HCC381	Pressure ulcer of skin with full thickness skin loss	HCC382, HCC383
HCC382	Pressure ulcer of skin with partial thickness skin loss	HCC383
HCC383	Chronic ulcer of skin, except pressure, not specified as through to bone or muscle	N/A
HCC385	Severe skin burn	N/A
HCC387	Pemphigus, pemphigoid, and other specified autoimmune skin disorders	N/A
HCC397	Major head injury with loss of consciousness (> 1 hour)	HCC202, HCC398, HCC399
HCC398	Major head injury with loss of consciousness (< 1 hour or unspecified)	HCC202, HCC399
HCC399	Major head injury without loss of consciousness	N/A
HCC401	Vertebral fractures without spinal cord injury	N/A
HCC402	Hip fracture/dislocation	N/A

HCC	HCC Description	Competing Events
HCC405	Traumatic amputations and complications	HCC409
HCC409	Amputation status, lower limb/amputation complications	N/A
HCC454	Stem cell, including bone marrow, transplant status/complications	N/A
HCC463	Artificial openings for feeding or elimination	N/A

1.1. Summary statistics from All of Us survey data

These data represent included survey respondents with at least one self-reported health condition that mapped to a Hierarchical Condition Category (HCC) in Version 28 of the Medicare Advantage risk adjustment algorithm. Only sets of co-occurring HCCs with more than 21 respondents (which also excludes several sets with more than 20 respondents) were used in any summary tables to comply with the All of Us Data and Statistics Dissemination Policy.

Table 2: **Included surveys from All of Us**

All of Us Survey Title	Number of Unique Respondents
Overall Health	156
Personal and Family Health History	15692

Table 3: **Number of respondents by HCC with available survey questions**

Version 28 HCC	Number of Respondents
HCC1	546
HCC20	339
HCC21	5775
HCC22	260
HCC23	6772
HCC35	156
HCC38	2114
HCC51	4283
HCC62	156
HCC64	100
HCC65	546
HCC77	156
HCC78	511
HCC93	1131

Version 28 HCC	Number of Respondents
HCC109	1018
HCC155	1467
HCC182	47
HCC221	156
HCC226	73
HCC228	399
HCC238	1635
HCC249	373
HCC264	62
HCC267	123
HCC276	156
HCC280	273
HCC300	587
HCC327	31
HCC328	110
HCC398	808

Table 4: **Distribution of HCC-relevant conditions per person**

Number of HCC-Relevant Conditions Per Person	Number of Respondents
1	5636
2	6657
3	3163
4	236
5	156

The following tables show respondent information using categories provided by All of Us. Not all respondents responded to all questions so counts may vary. The rows in each table correspond to available fields in All of Us. These characteristics are not displayed jointly because that would result in sample sizes too small to comply with the All of Us Data and Statistics Dissemination Policy.

Table 5: **Respondent age at survey response**

Age Group (years)	Number of Respondents	Percent of Respondents (%)
[65-70)	5088	32
[70-75)	5767	36
[75-80)	3383	21
[80-85)	1362	9
[85-90)	250	2

Table 6: **Respondents' self-reported sex at birth**

Self-Reported Sex at Birth	Number of Respondents	Percent of Respondents (%)
Female	8461	53
Male	6975	44
No matching concept	269	2
Not male, not female, prefer not to answer, or skipped	143	1

Table 7: **Respondents' self-reported ethnicity**

Self-Reported Ethnicity	Number of Respondents	Percent of Respondents (%)
Hispanic or Latino	633	4
Not Hispanic or Latino	14442	91
Prefer not to answer	34	0
Skipped	639	4
Neither	100	1

Table 8: **Respondents' self-reported race**

Self-Reported Race	Number of Respondents	Percent of Respondents (%)
Asian	260	2
Black or African American	752	5
White	13347	84
Another single population	49	0
More than one population	129	1
I prefer not to answer	34	0
Skipped	639	4
None indicated	538	3
None of these	100	1

2. Additional simulation results

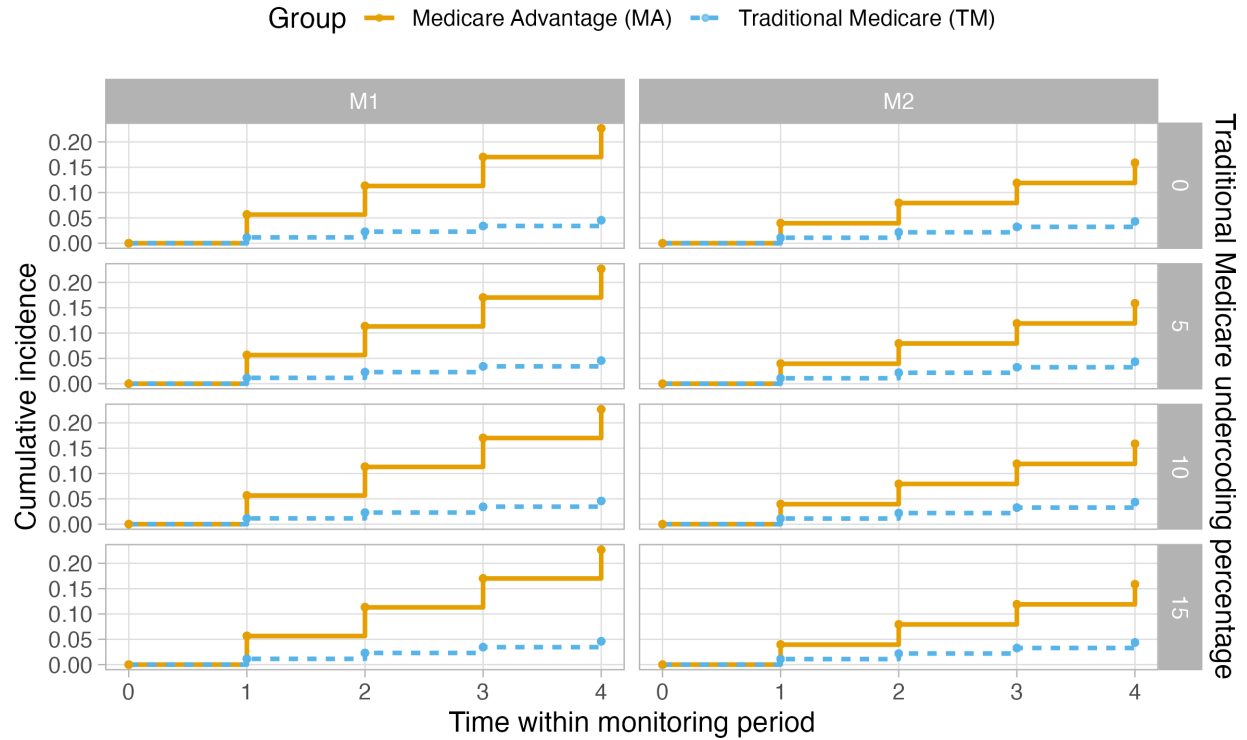


Figure 1: **Cumulative incidence functions for the Specified Heart Arrhythmias Hierarchical Condition Category (HCC) in simulated Medicare Advantage (MA) and Traditional Medicare (TM) groups: 25% any-available MA upcoding.** Specified heart arrhythmias corresponds to HCC238, which does not have any competing events. For this HCC, 25% of any-available individuals in the MA group are upcoded and 5% of any-available individuals in the TM comparison group are upcoded. Given the large sample size, confidence intervals are very narrow and are therefore omitted as they cannot be distinguished visually.

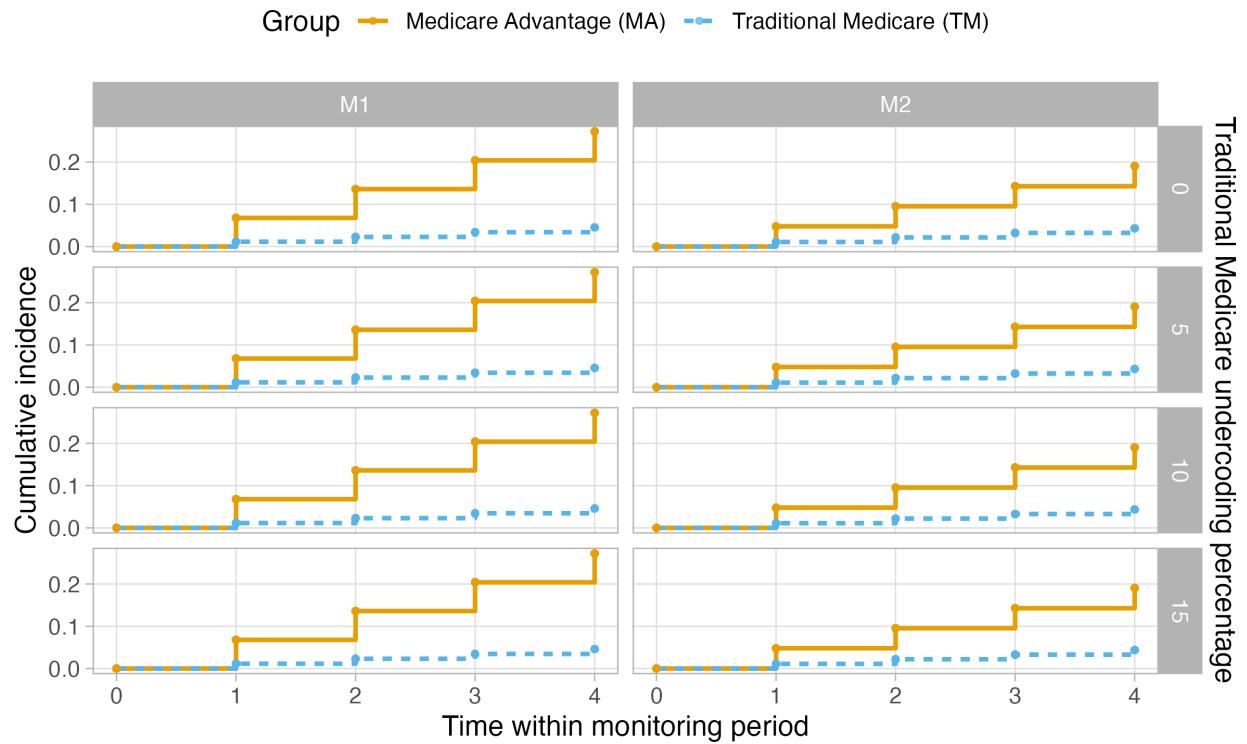


Figure 2: **Cumulative incidence functions for the Specified Heart Arrhythmias Hierarchical Condition Category (HCC) in simulated Medicare Advantage (MA) and Traditional Medicare (TM) groups: 30% any-available MA upcoding.** Specified heart arrhythmias corresponds to HCC238, which does not have any competing events. For this HCC, 30% of any-available individuals in the MA group are upcoded and 5% of any-available individuals in the TM comparison group are upcoded. Given the large sample size, confidence intervals are very narrow and are therefore omitted as they cannot be distinguished visually.

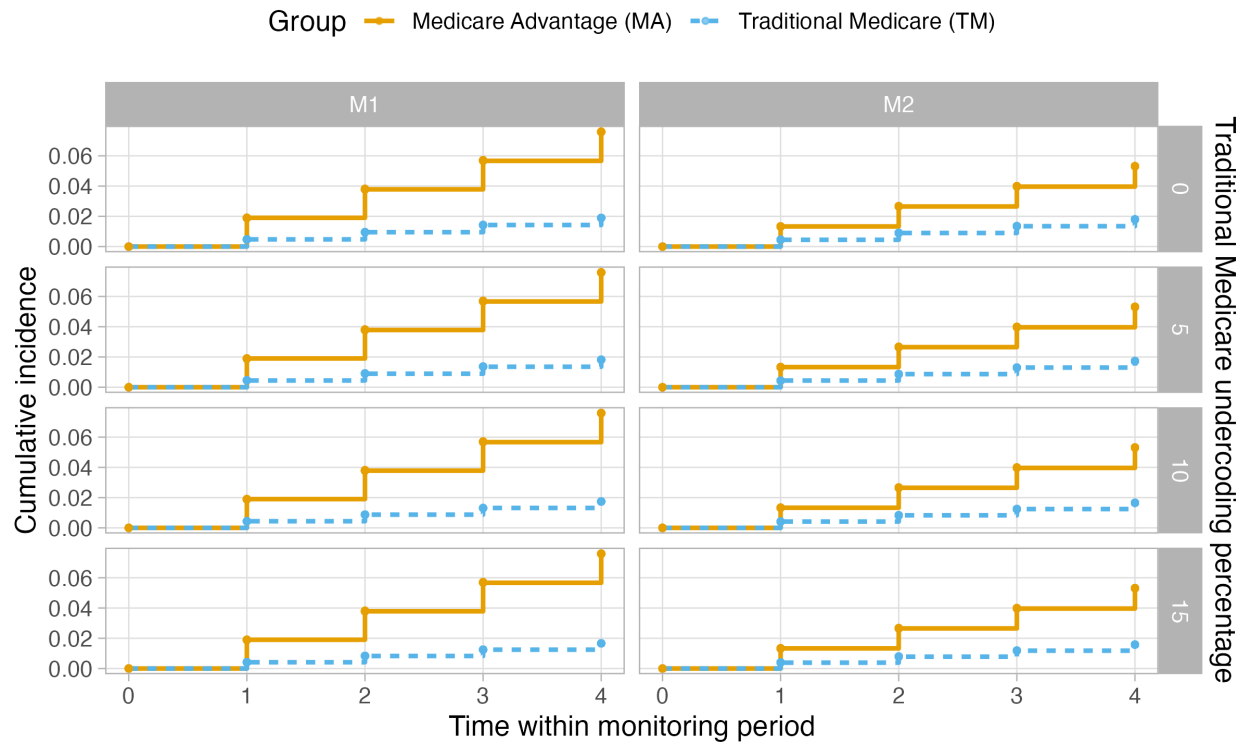


Figure 3: **Cumulative incidence functions for the ‘Dementia, Severe’ Hierarchical Condition Category (HCC) in simulated Medicare Advantage (MA) and Traditional Medicare (TM) groups: 20% lower severity MA upcoding.** ‘Dementia, Severe’ corresponds to HCC125, which has lower severity HCCs ‘Dementia, Moderate’ (HCC126) and ‘Dementia, Mild or Unspecified’ (HCC127). 20% of any individuals in the MA group who were previously coded with either HCC126 or HCC127 are upcoded and 5% of individuals in the TM comparison group are upcoded similarly. Given the large sample size, confidence intervals are very narrow and are therefore omitted as they cannot be distinguished visually.

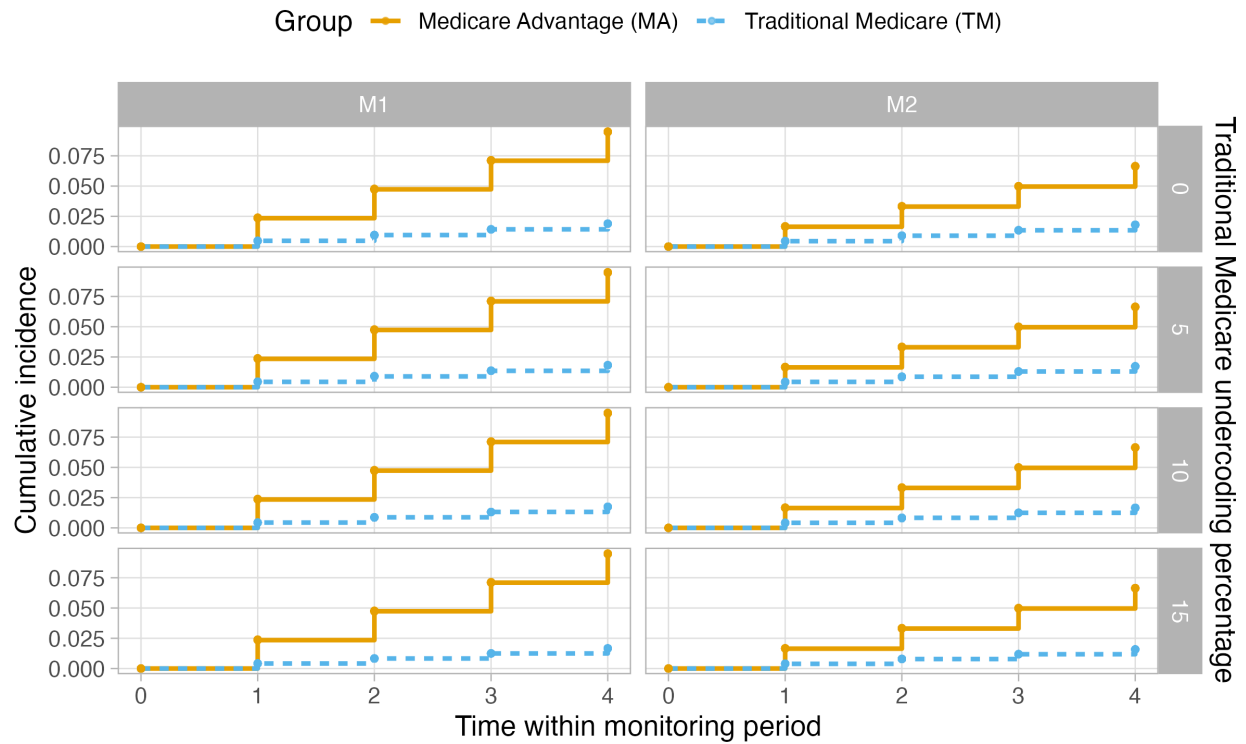


Figure 4: **Cumulative incidence functions for the ‘Dementia, Severe’ Hierarchical Condition Category (HCC) in simulated Medicare Advantage (MA) and Traditional Medicare (TM) groups: 25% lower severity MA upcoding.** ‘Dementia, Severe’ corresponds to HCC125, which has lower severity HCCs ‘Dementia, Moderate’ (HCC126) and ‘Dementia, Mild or Unspecified’ (HCC127). 25% of any individuals in the MA group who were previously coded with either HCC126 or HCC127 are upcoded and 5% of individuals in the TM comparison group are upcoded similarly. Given the large sample size, confidence intervals are very narrow and are therefore omitted as they cannot be distinguished visually.

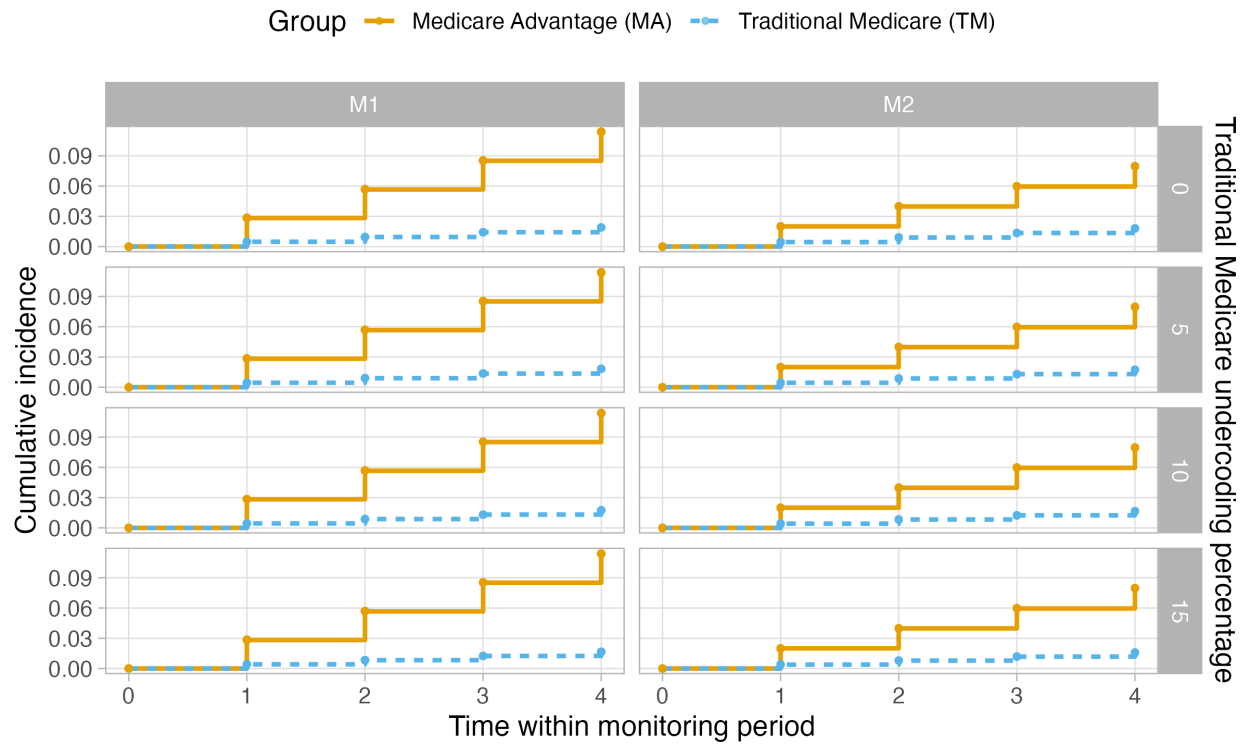


Figure 5: **Cumulative incidence functions for the ‘Dementia, Severe’ Hierarchical Condition Category (HCC) in simulated Medicare Advantage (MA) and Traditional Medicare (TM) groups: 30% lower severity MA upcoding.** ‘Dementia, Severe’ corresponds to HCC125, which has lower severity HCCs ‘Dementia, Moderate’ (HCC126) and ‘Dementia, Mild or Unspecified’ (HCC127). 30% of any individuals in the MA group who were previously coded with either HCC126 or HCC127 are upcoded and 5% of individuals in the TM comparison group are upcoded similarly. Given the large sample size, confidence intervals are very narrow and are therefore omitted as they cannot be distinguished visually.

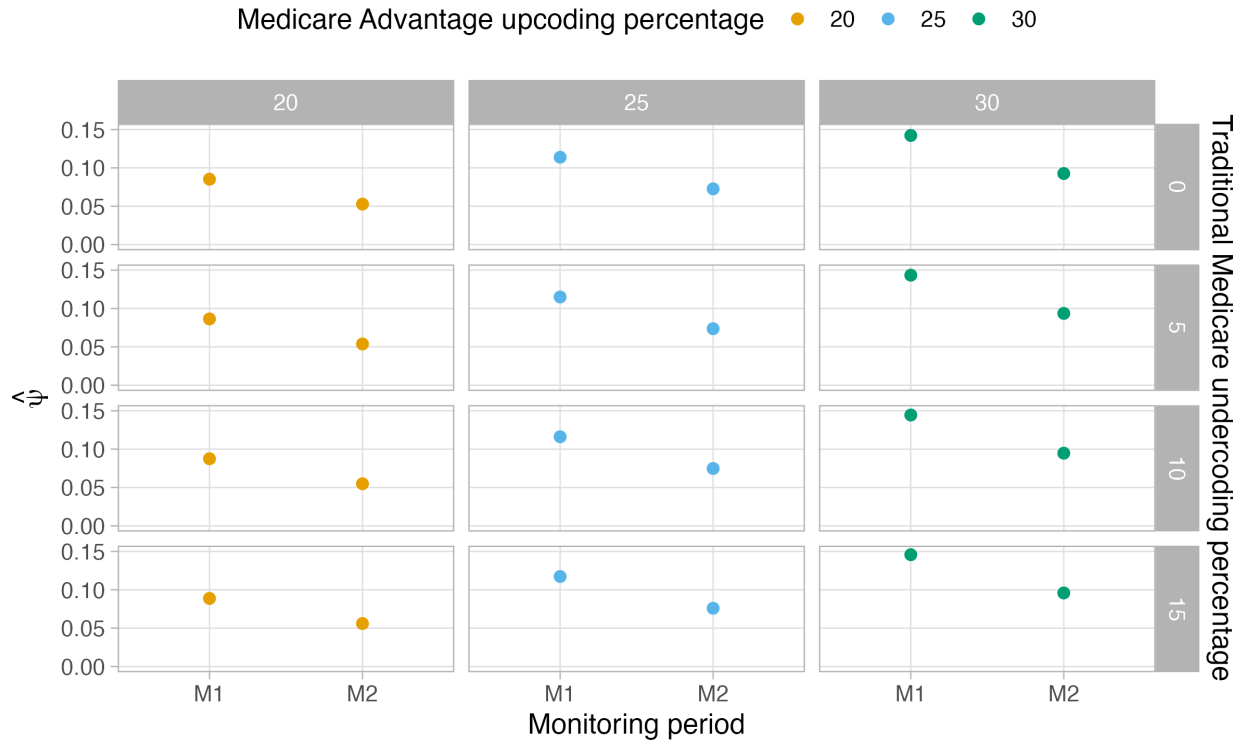


Figure 6: **Within-monitoring period ψ estimates for the ‘Dementia, Severe’ Hierarchical Condition Category (HCC) in simulated Medicare Advantage (MA) and Traditional Medicare (TM) groups.** ‘Dementia, Severe’ corresponds to HCC125, which has lower severity HCCs ‘Dementia, Moderate’ (HCC126) and ‘Dementia, Mild or Unspecified’ (HCC127). For this HCC, individuals in the MA group who were previously coded with either HCC126 or HCC127 are upcoded to varying degrees and individuals in the TM comparison group are upcoded 5% similarly. Given the large sample size, confidence intervals are very narrow and are therefore omitted as they cannot be distinguished visually.

3. Data availability statement

The simulated data for this study can be generated using code in the project repository: https://github.com/StanfordHPDS/tte_estimation_medicare. Additional upcoding, undercoding, and baseline data can be simulated using the upcoding package (<https://github.com/StanfordHPDS/upcoding>). In line with program policies, non-summarized version 7 All of Us survey data used to derive baseline co-occurring HCCs can be accessed by authorized users via the All of Us Researcher Workbench only (<https://www.researchallofus.org/data-tools/workbench/>) and requires Registered Tier access. Code that was used for this project in the Researcher Workbench is in the above project repository.

References

- Centers for Medicare and Medicaid Services. 2024 MA Advance Notice.; 2023. <https://www.cms.gov/files/document/2024-advance-notice-pdf.pdf>

2. Centers for Medicare and Medicaid Services. 2024 model software/ICD-10 mappings, 2024 midyear/final model software. Published online 2024. <https://www.cms.gov/medicare/health-plans/medicareadvtspecratestats/risk-adjustors/2024-model-software/icd-10-mappings>